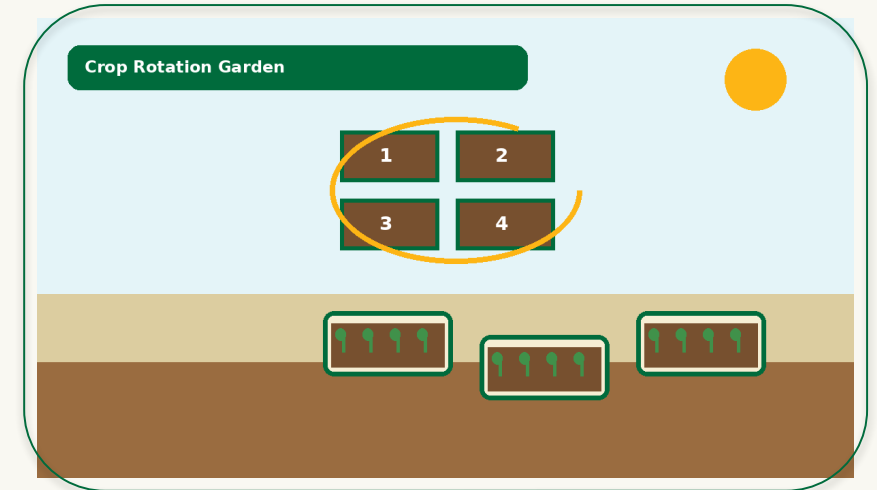


Crop Rotation Garden

Planning Crops Across Seasons

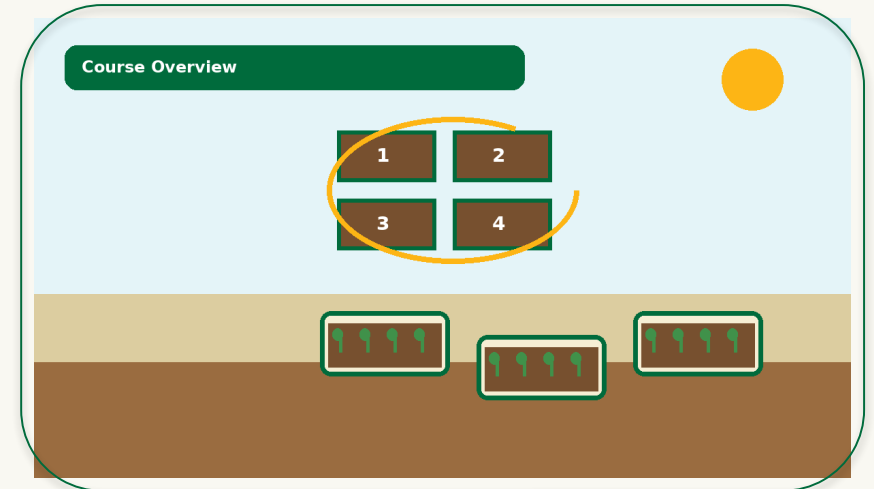
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Learning progression: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system and its educational purpose; teach all ten technical topics at Primary 4-6 level; guide safe practical work and measurement; connect science, mathematics and agriculture; maintain useful records; help pupils interpret observations without overclaiming.
- Core materials: teacher guide, system diagram, child-safe tools, labels, clipboards, rulers/tapes, graduated containers or measuring devices where relevant, observation/data sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson, measurement or investigation plan.



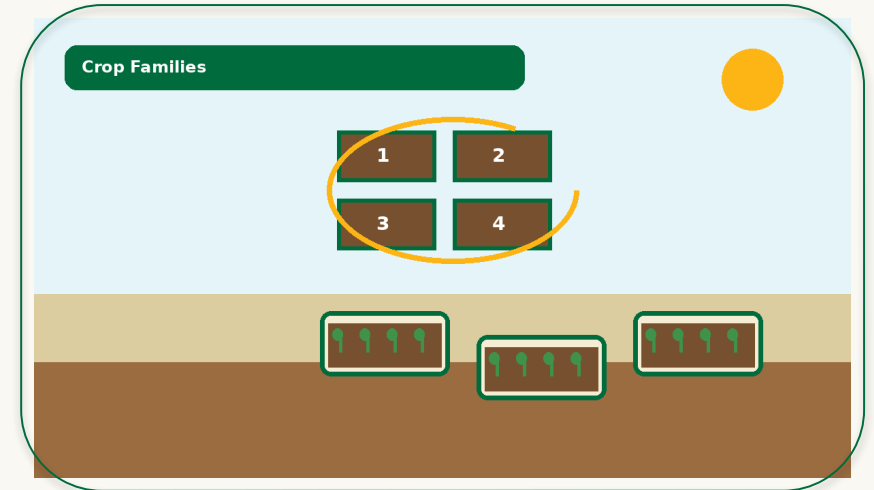
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Crop Families

Teacher knowledge and practical application

- Introduce crop families as groups of related plants that can share biological characteristics.
- Use a small set of locally relevant examples rather than requiring extensive taxonomy.
- Explain that family grouping can help organize rotation planning.
- Use crop cards or a garden map for sorting.
- Verify crop-family information before teaching.



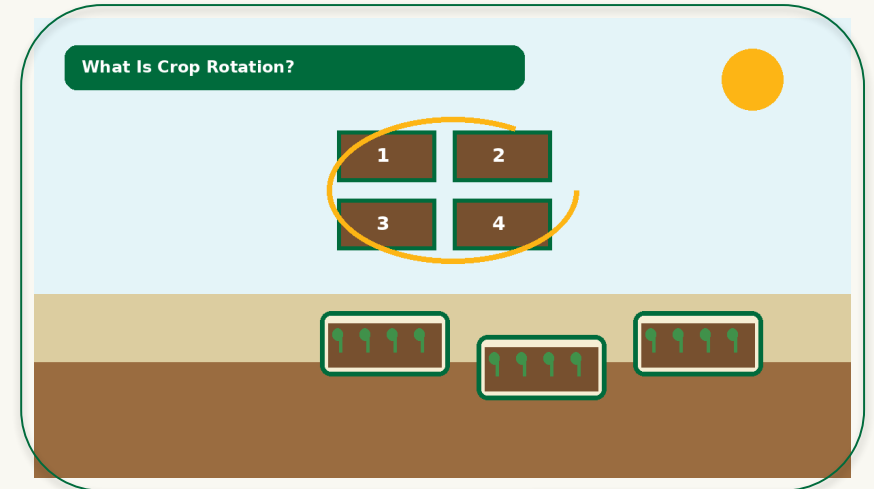
TEACHER PRACTICE

- Prepare crop-family cards for the crops used by the school.

What Is Crop Rotation?

Teacher knowledge and practical application

- Define crop rotation as planned change in the crops or crop groups grown in a space over successive growing periods.
- Contrast rotation with repeatedly growing the same crop in one bed.
- Explain that rotation is one management tool among several.
- Avoid promising that rotation alone prevents all pest or fertility problems.
- Connect the concept to labeled garden blocks.



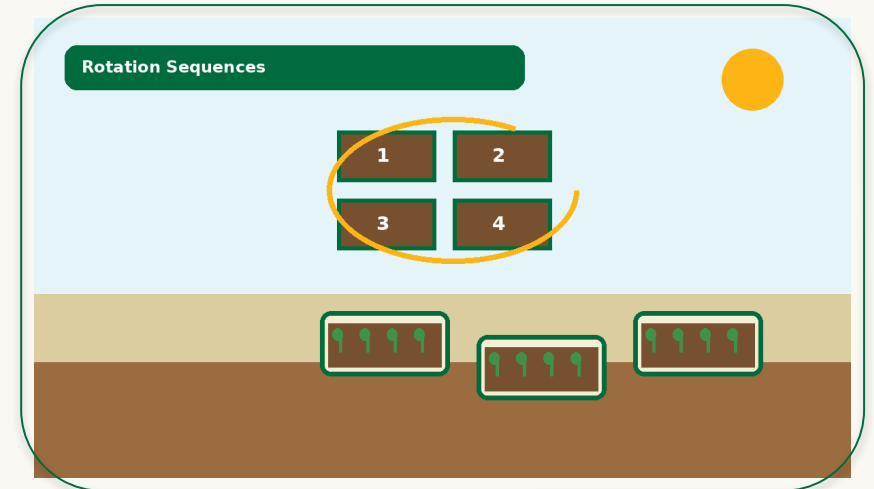
TEACHER PRACTICE

- Trace one bed across three planned crop periods.

Rotation Sequences

Teacher knowledge and practical application

- Build a simple sequence using crop families or functional groups appropriate to the school plan.
- Show the sequence with arrows or a seasonal calendar.
- Consider crop duration and school term timing.
- Avoid sequences that are agronomically unsuitable for local conditions.
- Record what actually gets planted, not only the original plan.



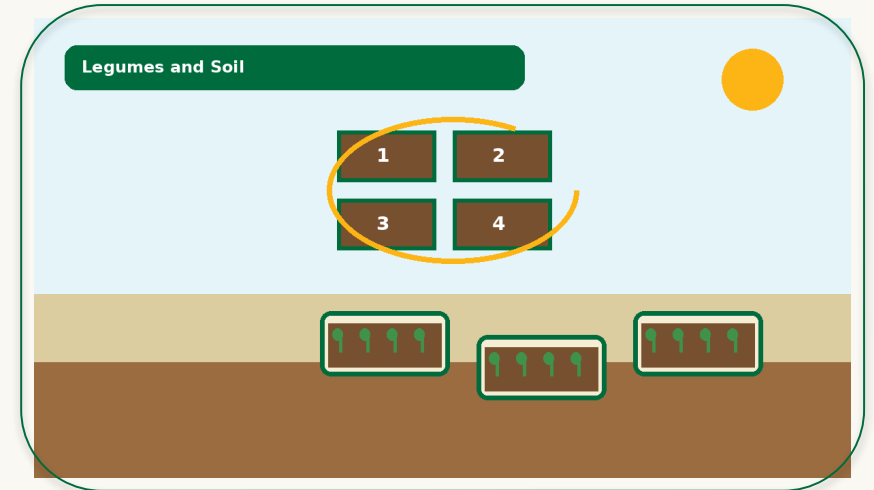
TEACHER PRACTICE

- Create a three-stage rotation sequence.

Legumes and Soil

Teacher knowledge and practical application

- Introduce legumes such as beans or cowpea as crops associated with nitrogen-fixing bacteria in root nodules.
- Explain that nitrogen benefits depend on species, nodulation and management; do not say legumes automatically 'add nitrogen' to every following crop.
- Observe roots only from planned harvested samples, not by uprooting active plots.
- Connect residues and soil management to the rotation discussion.
- Use simple language while keeping the science accurate.



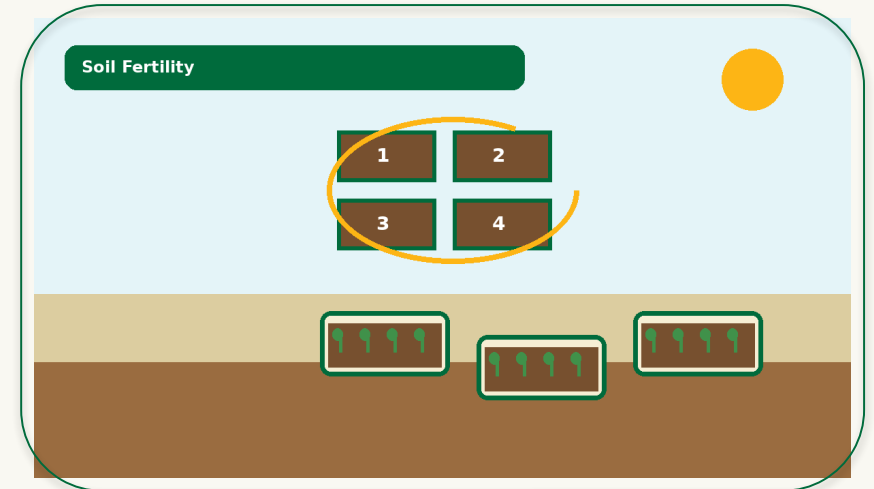
TEACHER PRACTICE

- Prepare a legume role-in-rotation explanation.

Soil Fertility

Teacher knowledge and practical application

- Explain that different crops remove and return nutrients in different ways.
- Use compost and approved fertility management alongside rotation.
- Observe soil and crop performance across seasons.
- Avoid diagnosing nutrient status from crop appearance alone.
- Use records to decide when soil testing or additional guidance is needed.



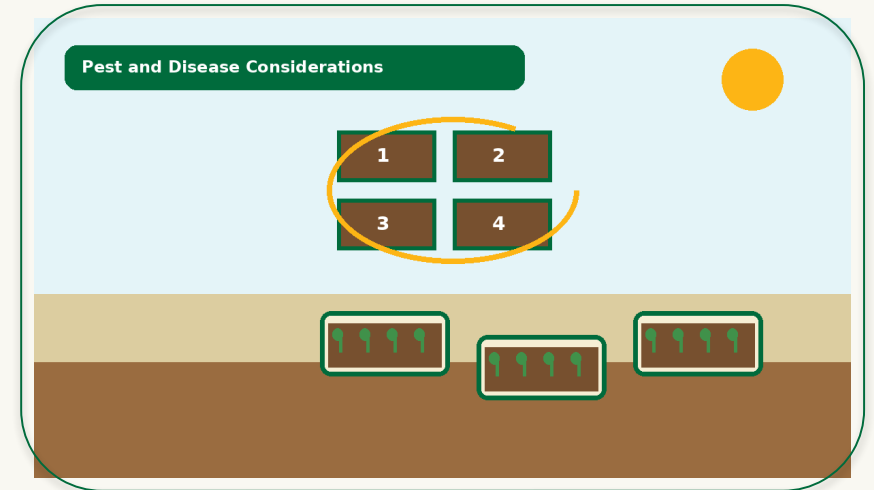
TEACHER PRACTICE

- Add fertility notes to the rotation record.

Pest and Disease Considerations

Teacher knowledge and practical application

- Explain that repeatedly growing closely related crops can sometimes favor persistence of certain crop-specific pests or diseases.
- Rotation can reduce some risks but is not a complete pest-management strategy.
- Identify volunteer plants and crop residues that may affect rotation.
- Use approved integrated pest-management procedures.
- Pupils observe and record; pesticide handling is excluded.



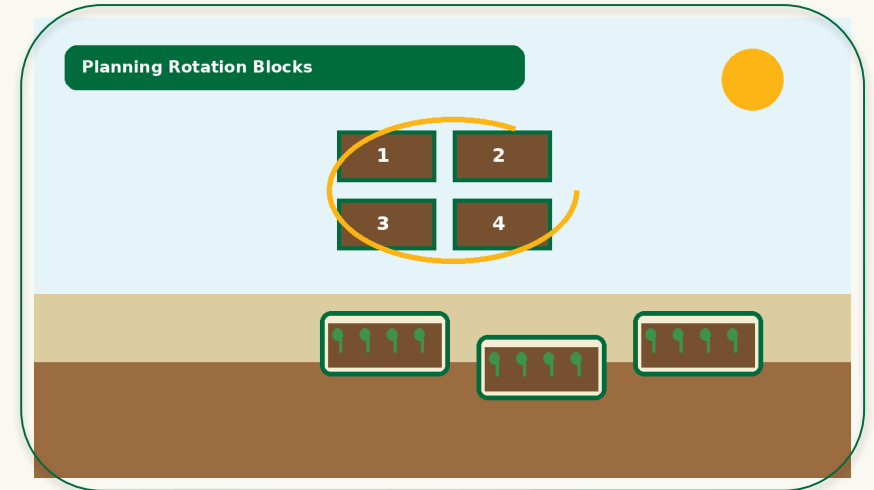
TEACHER PRACTICE

- Prepare a simple rotation-and-pest scenario discussion.

Planning Rotation Blocks

Teacher knowledge and practical application

- Divide the garden into clearly labeled blocks or beds.
- Keep block identity consistent across seasons.
- Assign crop groups according to the planned sequence.
- Consider sunlight, irrigation and crop duration.
- Use a map so pupils can follow movement between blocks.



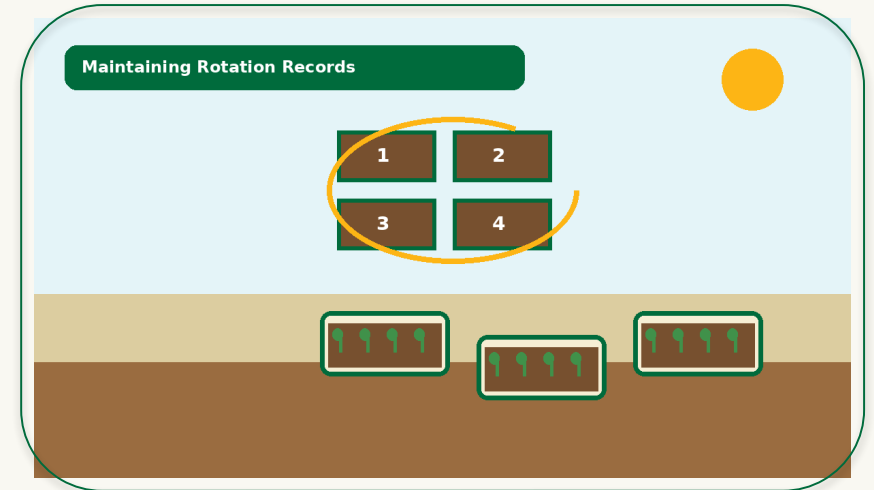
TEACHER PRACTICE

- Draw a four-block rotation map.

Maintaining Rotation Records

Teacher knowledge and practical application

- Record block, crop, family/group, planting date and harvest/end date.
- Note major pest, disease or fertility observations.
- Keep records across school years where possible.
- Update the map when plans change.
- Use records before deciding what should be planted next.



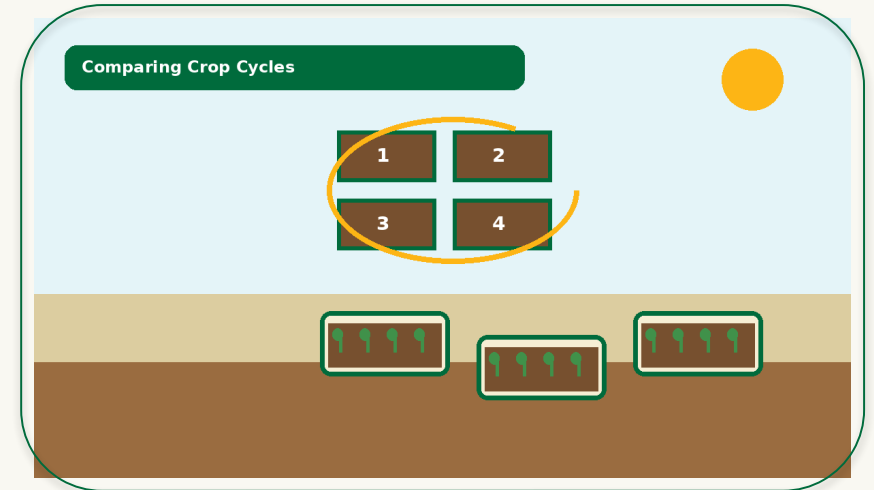
TEACHER PRACTICE

- Create a rotation-history table.

Comparing Crop Cycles

Teacher knowledge and practical application

- Compare crop duration, growth, harvest and management demands across rotation stages.
- Use the same measurement definitions when comparing similar crops.
- Recognize that seasons and weather can also influence results.
- Do not attribute every difference to rotation.
- Use comparison to generate questions for future cycles.



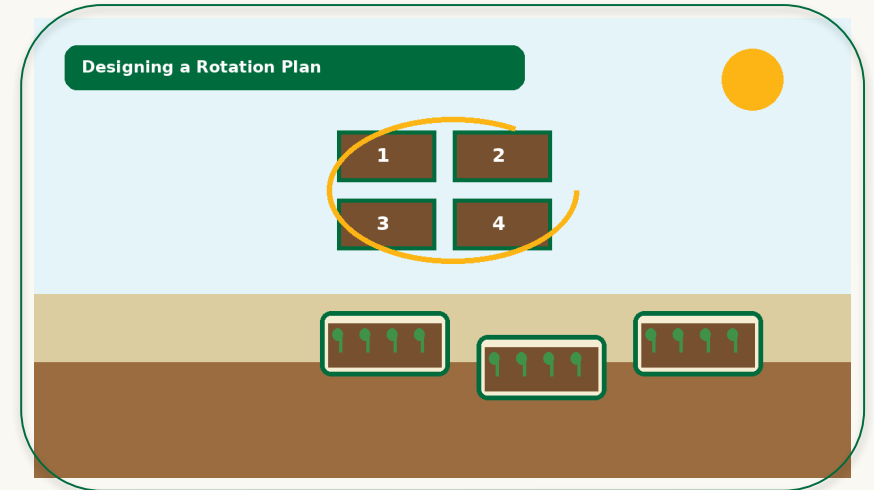
TEACHER PRACTICE

- Compare two completed crop cycles using records.

Designing a Rotation Plan

Teacher knowledge and practical application

- Start with available beds, target crops and expected planting windows.
- Group crops appropriately and assign a sequence.
- Check that the same family is not unintentionally repeated where avoidance is a goal.
- Include compost/fertility and maintenance needs.
- Review and revise the plan as actual harvest dates change.



TEACHER PRACTICE

- Completion task: design a one-year school crop-rotation plan.